Session 5: PolymorphismTopics:Compile-time (overloading) vs Runtime (overriding).Example: uploadContent() works differently for YouTube vs Instagram.

Tasks

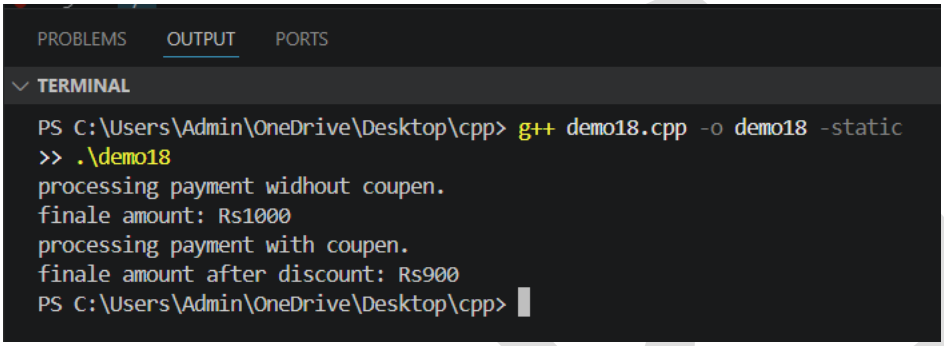
1) **Create a cpp class called PaymentProcessor with two overloaded methods processPayment(): one that takes only an amount, and one that takes amount and a coupon code. Print which version is called and the final amount in each case.**

Code-

```
#include<iostream>
#include <string>
using namespace std;


class PaymentProcessor{

   public:
   void  processPayment(double amount){
      cout<<"processing payment widhout coupen."<<endl;
        cout<<"finale amount: Rs"<<amount<<endl;

   }

   void processPayment(double amount,string coupen){
         double finalAmount = amount - (amount * 0.10);

         cout<<"processing payment with coupen."<<endl;
         cout<<"finale amount after discount: Rs"<<finalAmount<<endl;

   }
};

   int main(){
      PaymentProcessor p1;
```

```
    p1.processPayment(1000);

    p1.processPayment(1000, "SAVE10");


    return 0;
}
```

Output:



2) Build two classes, InstagramUploader and YouTubeUploader, each with a method uploadContent(). Both should extend a base class SocialMediaUploader and override uploadContent() to print a message showing how uploading works differently for Instagram and YouTube.

Code-

```
#include<iostream>
using namespace std;


class SocialMediaUploader{
   private:

  virtual void uploadContent(){
      cout<<" we uplodated content social media every day:"<<endl;

   }

};
class InstagramUploader:public SocialMediaUploader{
   public:
   void uploadContent()override{
      cout<<"we uplodated photos and reel natrual beauty prodects on insta:"<<endl;

   }

};
class YouTubeUploader:public SocialMediaUploader{
   public:
      void uploadContent()override{
         cout<<"we uplodated the  game title,GTA ,videos in eavry weak:"<<endl;

      }
};
int main(){
   InstagramUploader i1;
   YouTubeUploader yt1;
```

```
    cout<<"instagram uplodated"<<endl;
    i1.uploadContent();
    cout<<"\nyoutube updated"<<endl;
    yt1.uploadContent();

    return 0;

}
```

Output:

```
PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo19.cpp -o demo19 -static
>> .\demo19
instagram uplodated
we uplodated photos and reel natrual beauty prodects on insta:

youtube updated
we uplodated the  game title,GTA ,videos in eavry weak:
PS C:\Users\Admin\OneDrive\Desktop\cpp>
```

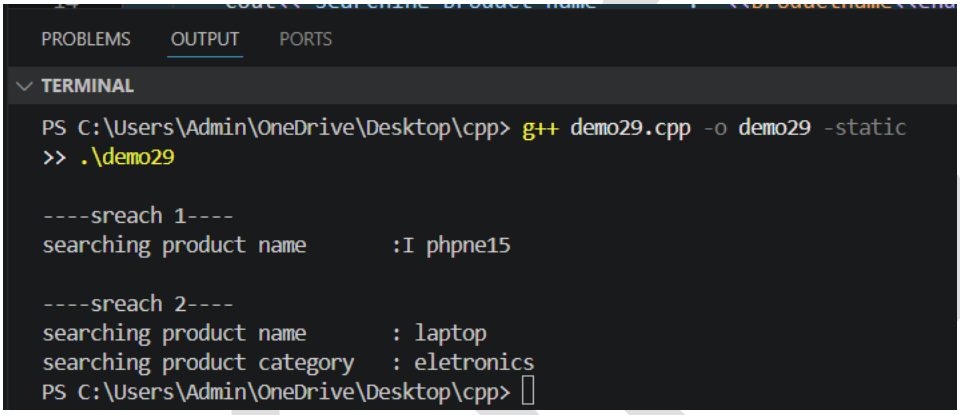
3) **Write a function in c++ that simulates a Flipkart-style search: overload a method searchProduct() to allow searching by product name or by product name and category. Demonstrate both usages with sample data.**

Code-

```
#include<iostream>
using namespace std;

class producsearch {

   public:

   void searchProduct(string productname){
      cout<<"searching product name      :"<<productname<<endl;

   }
   void sreachProduct(string productname, string catagory){
      cout<<"searching product name      : "<<productname<<endl;
      cout<<"searching product category   : "<<catagory<<endl;
   }
};
int main(){
   producsearch search;

   cout<<"\n----sreach 1----"<<endl;
   search.searchProduct("I phpne15");
```

```
    cout<<"\n----sreach 2----"<<endl;
    search.sreachProduct("laptop","eletronics");

    return 0;

}
```

Output:



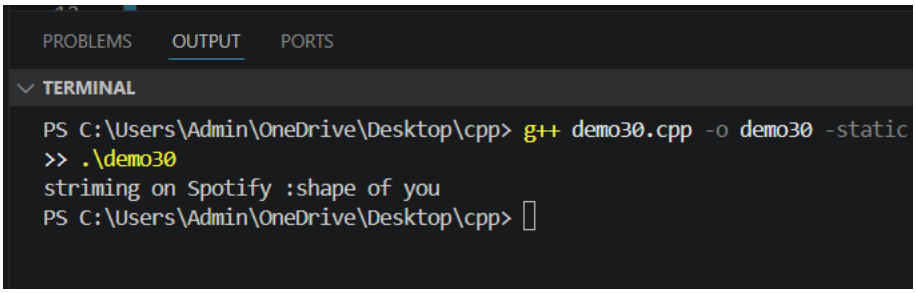
4) **Given this code: class MusicPlayer { void play(String song) { System.out.println("Playing: " + song); } } class SpotifyPlayer extends MusicPlayer { void play(String song) { System.out.println("Streaming on Spotify: " + song); } } — Create an object of type MusicPlayer but assign it a SpotifyPlayer instance, then call play(). Explain the output.

Hint: This tests runtime polymorphism (overriding) and dynamic method dispatch.**

Code-

```
#include<iostream>
#include<string>
using namespace std;

class MusicPlayer{

   public:
   virtual void play(string song){
      cout<<"playing :"<<song<<endl;

   }

};
class SpotifyPlayer:public MusicPlayer{
   public:
   void play(string song)override{
      cout<<"striming on Spotify :"<<song<<endl;
   }
};
int main(){
   MusicPlayer *m1=new SpotifyPlayer();
   m1->play("shape of you");
   delete m1;
   return 0;
}
```

Output:


PROBLEMS
OUTPUT
PORTS
TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo30.cpp -o demo30 -static
>> .\demo30
striming on Spotify :shape of you
PS C:\Users\Admin\OneDrive\Desktop\cpp>